

Question 1

CP - Create a program to check if a number taken from the user is a Harshad Number.

Hint =>

1. A Harshad number is an integer that is divisible by the sum of its digits. For example, 21 which is perfectly divided by 3 (sum of digits: $2 + 1$).
2. Get an integer input for the number variable.
3. Create an integer variable sum with an initial value of 0.
4. Create a while loop to access each digit of the number.
5. Inside the loop, add each digit of the number to the sum.
6. Check if the number is perfectly divisible by the sum.
7. If the number is divisible by the sum, print Harshad Number. Otherwise, print Not a Harshad Number.

Question 2

CP - Write a LeapYear program that takes a year as input and outputs the Year is a Leap Year or not a Leap Year.

Hint =>

1. The LeapYear program only works for year ≥ 1582 , corresponding to a year in the Gregorian calendar. So ensure to check for the same.
2. Further, the Leap Year is a Year divisible by 4 and not 100 unless it is divisible by 400. E.g. 1800 is not a Leap Year and 2000 is a Leap Year.
3. Write code having multiple if else statements based on conditions provided above

Question 3

CP - Write a program DayOfWeek that takes a date as input and prints the day of the week that the date falls on.

Hint =>

1. The program should take three command-line arguments: m (month), d (day), and y (year). For m use 1 for January, 2 for February, and so forth.

2. For output print 0 for Sunday, 1 for Monday, 2 for Tuesday, and so forth.

3. Use the following formulas, for the Gregorian calendar (where / denotes integer division):

$$y_0 = y - (14 - m) / 12$$

$$x = y_0 + y_0/4 - y_0/100 + y_0/400$$

$$m_0 = m + 12 \times ((14 - m) / 12) - 2$$

$$d_0 = (d + x + 31m_0 / 12) \bmod 7$$

Question 4

CP - Write a Program to check if the given number is a prime number or not

Hint =>

1. A number that can be divided exactly only by itself and 1 are Prime Numbers,
2. Prime Numbers checks are done for numbers greater than 1
3. Loop through all the numbers from 2 to the user input number and check if the remainder is zero. If the remainder is zero break out from the loop as the number is divisible by some other number and is not a prime number.
4. Use the isPrime boolean variable to store the result and finally display the result

Question 5

CP - Create a program to count the number of digits in an integer.

Hint =>

1. Get an integer input for the number variable.
2. Create an integer variable count with the value 0.
3. Use a loop to iterate until the number is not equal to 0.
4. Remove the last digit from the number in each iteration
5. Increase the count by 1 in each iteration.
6. Finally, display the count to show the number of digits

